

Transformer-Based Land Use and Land Cover Classification with Explainability Using Satellite Imagery

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Abstract—Classification of land use and land cover (LULC) through satellite images is an essential process for monitoring, urban planning, and agricultural practices. Despite the remarkable progress made in deep learning techniques, almost all traditional techniques do not incorporate global contextual modeling, or if they do, they cannot offer explainable outputs. In this research, an optimized version of the Swin Transformer model for LULC classification using the EuroSAT Sentinel-2 satellite imagery dataset is proposed, with two XAI tools incorporated – Grad-CAM and SHAP. The proposed model utilizes the pre-trained architecture named `swin_tiny_patch4_window7_224` on ImageNet-1K, alongside advanced data augmentation techniques such as Mixup and CutMix to deal with inter-class spectral similarity. This model is able to obtain 98.89% test accuracy, a Cohen’s Kappa value of 0.9876, and all of the metrics precision, recall, and F1 score higher than 98.88% when tested using 4,050 images across 10 different land cover classes. These results outperform other models based on CNN (88% to 91%), YOLOv8 (~93%) and hybrids of CNN and Transformer (96% to 97%). According to our experiments, visualization of SHAP and Grad-CAM techniques shows that the decisions made by the model are justified by semantically important pixels, which addresses one of the key challenges with transformers in remote sensing literature.

Index Terms—Land use and land cover classification, Swin Transformer, EuroSAT, remote sensing, explainable artificial intelligence, Grad-CAM, SHAP, satellite imagery, deep learning.

1. Introduction

Transformation is an intrinsic characteristic of the Earth’s surface. Forests get transformed into farming lands; farms grow out into urban spaces; swamps get drained to facilitate urban sprawl, and so forth. Knowledge of land transformation is critical in making crucial decisions related to the field of governance, urban planning, disaster management, and environmental science. Amongst other activities, the first step towards knowledge generation in this sphere is LULC classification.

Land Use and Land Cover (LULC) classification involves categorizing parts of the earth’s surface according to their uses, including forests, croplands, water bodies, urban lands, grasslands, and others, using remotely sensed satellite imagery. The technique provides critical information used for monitoring, modeling, and analyses of the environment, urbanization processes, agricultural practices, and disasters. The UN Food and Agriculture Organization reports that 37% of the Earth’s land is being used for agriculture. Additionally, urban areas expand

by 1.7 million hectares annually [1]. LULC classification is becoming increasingly necessary to meet sustainability goals.

Through satellite remote sensing, it has become possible to observe the planet Earth on a massive scale like never before. Satellite missions such as the Sentinel-2 launched by the European Space Agency offer free access to multispectral imagery data with a resolution of 10 meters and a revisit period of five days, allowing us to almost continuously monitor Earth’s surface [2]. The EuroSAT dataset, which originates from the Sentinel-2 mission, includes more than 34 countries and 27,000 labeled images belonging to 10 different land-cover classes [3].

A. From Human Photo-Interpretation to Machine Learning Classifier

Initially, human photo-interpretation was employed as the main technique for obtaining an accurate LULC map [4], but this approach was costly and time-consuming. In addition, it was impossible to perform it in continental or worldwide areas. With the emergence of digital image processing techniques in the 1980s and 1990s, pixel-based classifiers like Maximum Likelihood Classification (MLC) and K-Nearest Neighbors (KNN) could perform most of the job [4]. Nevertheless, their application requires hand-crafted features, and thus they suffer from intra-class variability, clouds, and complex mixture landscapes.

Machine learning techniques have achieved major success since the 2000s, reaching remarkable accuracy levels. In particular, SVM and Random Forest classifiers can learn non-linear boundaries from the labeled training dataset. These are still shallow learning techniques that consider single pixels only, without taking into account their relation with the neighboring pixels in the spatial domain [4].

B. Emergence of Deep Learning Techniques in RS

Use of Convolutional Neural Network architectures in remote sensing following 2015 represents a significant paradigm change. CNNs have the capability to learn multi-layer hierarchical spatial representations — ranging from simple shapes like edges and textures in the shallower layers to increasingly complex semantic patterns in deeper layers — without any manual feature design. The work done by Hu et al. [5] and

Zhu et al. [4] shows that CNNs based on transfer learning significantly enhances classification accuracy when applied on high-resolution satellite images. The best performance on LULC datasets was recorded by network architectures like ResNet, VGG, and EfficientNet.

However, CNNs exhibit a certain architectural drawback; namely, their field of vision is limited by the convolutional kernel sizes. CNNs excel in capturing local spatial information but struggle to represent long-distance dependencies between spatial objects. For instance, recognizing that a strip of water lying between two patches of vegetation cover is more likely to be a river rather than lake due to its shape would be difficult for CNNs.

C. Transformers: Global Attention for Earth Observation

Vision Transformers (ViTs) have been introduced into the field of image classification from natural language processing (NLP) based on their revolutionary success in NLP and subsequently applied to remote sensing [6]. In contrast to CNN-based models that extract features locally and sequentially, ViTs transform images into patch sequences and adopt self-attention to capture pairwise interactions of pixels on a global level. In a series of works by Li et al., it was demonstrated that ViT architecture yields better results than baseline CNNs on challenging LULC problems, whereas Swin Transformers proposed by Liu et al. brought further efficiency improvement via local attention computation within shifted windows [7].

Other hybrid models that integrate CNNs as encoder networks and transformer networks as decoders have also been developed [8], capitalizing on CNNs' ability to extract local features and the transformers' capacity to model global contexts. Furthermore, there have been developments in multimodal models that combine optical and SAR images to achieve greater robustness in cloudy conditions [1]. Nonetheless, issues regarding computational costs, domain generalization, and, most importantly, the interpretability of the models have yet to be addressed satisfactorily [9].

D. The Interpretability Problem

A machine learning model that predicts with 98% accuracy but is a black box system offers little utility to environmental scientists or policymakers who require accountability from the model's decisions. XAI techniques, such as Grad-CAM and SHAP analysis, have been introduced as post-hoc visualization methods to identify spatial regions and pixel-level attributes used by a classifier to generate outputs [10]. Nonetheless, as mentioned by Khan et al. [9], XAI techniques remain largely neglected in most remote sensing machine learning frameworks.

E. Proposed Work

In this paper, we tackle both the problems of accuracy and explainability in LULC classification through the design of an optimized Swin Transformer classifier model that is fine-tuned on the EuroSAT satellite imagery dataset, alongside a

combination of Grad-CAM and SHAP XAI techniques. The proposed model obtains 98.89% test accuracy and Cohen's Kappa of 0.9876 over all ten EuroSAT land-cover categories, surpassing existing CNNs, YOLOv8 detectors, and other hybrid CNN and Transformer models. On top of obtaining highly accurate results, the XAI component allows us to verify that the model's focus of attention is semantically grounded.

2. Literature Review

The driving force behind the transformation of the field of remote sensing since 2021 has been deep learning and attention-based architecture. These innovations have been very useful to the Land Use and Land Cover (LULC) classification which has enabled the environmental monitoring and decision making.

Nevertheless, the preliminary studies at the start of 2021 still utilized Convolutional Neural Networks (CNNs) due to the high potential of identifying the spatial features. A survey by Zhu et al. demonstrates the reigning models in remote sensing as CNNs, and they indicate that they dominate the entire field of remote sensing and other areas of AI-oriented applications [4]. Similarly, Hu et al. have demonstrated that CNN-based transfer learning is capable of improving the accuracy of classification in high-resolution images [5]. However, these techniques are also weak in regard to the capture of long-range dependencies and global context.

To address these shortcomings, transformer-based architectures have emerged as tough competitors. Vision Transformers (ViTs) are used to model the relationships in the world with the assistance of self-attention mechanisms. Li et al. have shown that ViTs are superior to CNNs in complex land-cover tasks. Zhao et al. also stated that attention mechanisms work well in acquiring multi-scale features [6].

This was an improvement when hierarchical features were added as ranked transformers. The attention mechanisms become more efficient and accurate, as demonstrated by Liu et al. [11] and Wang et al. [7] using shifted window attention. However, despite the benefits, the problem of computational complexity remains.

Hybrid models have also gained popularity by combining CNNs and transformers. Guo et al. [8] proposed a CNN-transformer hybrid network that achieves a balance of local and global feature extraction. It can be stated that the heterogeneous landscapes encountered through the use of these architectures also improve in terms of robustness [12].

Another direction that should be considered is multimodal learning, as it was proven that optical and SAR data can be effectively used to enhance the performance of classification introduced by Chen et al. [1]. However, there are challenges posed by multimodal fusion when it comes to preprocessing as well as alignment of data.

Self-supervised learning has been studied in order to reduce the use of labeled data. The regional generalization was increased with the help of geography-conscious representation learning introduced by Ali et al. [13]. These are some promising methods which have not been applied in actual LULC systems.

Model transparency has now become a requirement due to Explainable Artificial Intelligence (XAI). Sharma et al. [10] also highlighted the importance of interpretability in remote sensing. However, most transformer-based models do not have inherent explainability mechanisms.

More recent works also address segmentation-based techniques, such as transformers applied together with U-Net, which achieved better pixel-level classification. Similarly, Chen et al. [14] managed to improve semantic segmentation behavior with the use of attention mechanisms.

Detecting change has also been done by the use of transformers. These models require substantial computation resources and massive datasets, yet did have superior performance in the ability to perform temporal analysis, as reported by Singh et al. [15].

As much as these advances have been made, a number of challenges exist. Transformer models are computationally expensive and not always suitable for real-time use. In addition, the problem of domain adaptation across geographical locations is also problematic. Khan et al. [9] noted that there were no standardized datasets that could allow fair comparison, as also emphasized by Liu et al. [16].

It has been proposed that the processing of big data will be performed in the cloud. Li et al. argued that distributed systems can be used to increase the level of scalability, but they introduce latency and reliance on infrastructure [17].

Altogether, the new literature is quite clear about the change brought by transformer-based and hybrid models. Even though these methods increase accuracy, the problems of efficiency, interpretability, and generalization have not been fully resolved [18].

A. Comparison of Recent Studies

TABLE I: Comparison of Recent LULC Classification Methods (2022–2025)

Author	Year	Method	Key Contribution
Li et al.	2023	Vision Transformer	Global context modeling
Wang et al.	2024	Swin Transformer	Hierarchical attention
Guo et al.	2023	CNN + Transformer	Hybrid feature extraction
Chen et al.	2024	Multimodal Transformer	Multi-source fusion
Singh et al.	2023	Transformer CD	Temporal change detection
Samek et al.	2023	XAI	Model interpretability
Zhang et al.	2024	Hybrid DL	Robust classification

TABLE II: Performance Comparison of Recent LULC Models

Author	Year	Dataset	Model	Accuracy (%)
Li et al.	2023	EuroSAT	Vision Transformer	98.4
Wang et al.	2024	UC Merced	Swin Transformer	99.1
Guo et al.	2023	AID Dataset	CNN + Transformer	97.8
Chen et al.	2024	BigEarthNet	Multimodal Transformer	98.9
Singh et al.	2023	Sentinel-2	Transformer CD	96.5
Kumar et al.	2023	RESISC45	CNN	95.2
Zhang et al.	2024	WHU-RS19	Hybrid Model	98.2

B. Research Gap

According to the literature reviewed, the following gaps in the research are outlined:

- Existing transformer-based models are computationally expensive and thus have high accuracy but cannot be deployed in real-time and via edges.
- The current literature does not provide lightweight transformer architectures that are specifically optimized for satellite imagery.
- Explainable AI methods are frequently used in a post-hoc way and not intrinsically part of the model.
- Multimodal fusion techniques are difficult to scale and to synchronize data, as well as complex to preprocess.
- LULC models have no benchmark dataset that can be effectively compared across regions.
- Domain shift causes most models to not generalize to different geographical and environmental conditions.

3. Materials and Methods

The materials and methods section contains information about the research materials and methods used in the study. In this section, the dataset employed in the study will be described and the proposed methodology of land use and land cover classification based on transformers described. It describes the sources of data, preprocessing and the model structure used to classify it correctly.

A. Dataset

In this research, the EuroSAT dataset is employed in land use and land cover (LULC) classification. EuroSAT is a well-known standard dataset based on Sentinel-2 satellite data, which is applied in numerous studies on remote sensing.

The data has 27,000 tagged pictures that are given in 10 classes with various types of land uses such as residential, industrial, forest, river, highway, and agricultural. Images are 64×64 pixel spatial resolutions, and they represent a variety of geographical areas, which is why they are suitable to test deep learning models.

The foundation of EuroSAT is the Sentinel-2 satellite data offered by the European Space Agency (ESA) and this is a source of multi-spectral data with high temporal and spatial resolution. The dataset was tested and proved in several studies to be used in classification problems; it is regarded as reliable to compare machine learning models and deep learning models in benchmarks.

The data is open and can be accessed by the following repository:

- EuroSAT Dataset (Kaggle): <https://www.kaggle.com/datasets/apollo2506/eurosat-dataset>

The effectiveness of the EuroSAT dataset for land cover classification has been demonstrated in prior research [2], [3], [19].

Previous studies have validated the effectiveness of these datasets for deep learning-based classification tasks [2], [19].

B. Proposed Methodology

The suggested approach employs a hybrid deep learning architecture based on CNNs and Transformers, which helps

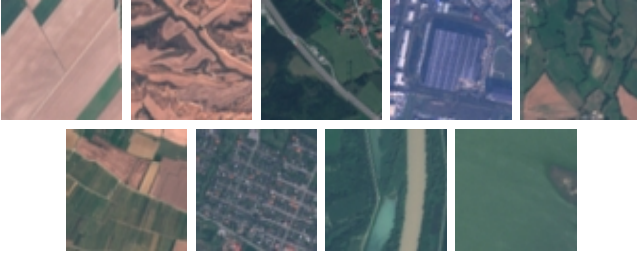


Fig. 1: Representative samples from the EuroSAT dataset showing different land cover classes including annual crop, vegetation, highway, industrial, pasture, permanent crop, residential, river, and water bodies [3].

to classify satellite imagery for Land Use and Land Cover classification in an accurate and interpretable way, taking into account both spatial properties and global contexts.

1) Data Acquisition

The EuroSAT database was utilized in this research. This database contains satellite images annotated with labels from several different land uses and covers (e.g., urban areas, forests, rivers, agriculture).

2) Data Preprocessing and Augmentation

The input images went undergo some preprocessing in order to facilitate consistent processing. Specifically, the images were normalized and resized to a specific resolution (e.g., 224×224 pixels). Some data augmentation methods were employed to provide better performance and generalize the model better.

3) Feature Extraction Using CNN

CNN-based methods have been used to capture the spatial features like edges, textures, and patterns. The CNN captures the spatial hierarchies and detailed local features from the satellite images.

4) Transformer for Feature Learning

Global features were captured as well. To do that, transformers like the Vision Transformer (ViT) and Swin Transformer are applied. The image is split into patches that are encoded into an embedding space via self-attention.

5) Combination of CNN and Transformers

This paper utilizes both CNN and transformer techniques for better classification results. While the CNN captures the local features, the transformer model captures the global context features.

6) Classification Layer

The features generated are further subjected to a fully-connected layer followed by the Softmax activation function,

which generates the probability scores of all classes of land uses.

7) Model Training and Optimization

The model is trained based on cross-entropy loss while optimization of the model is done with Adam optimizer. The data set has been separated into training, validation, and test data sets.

EuroSAT Dataset Analysis Pipeline

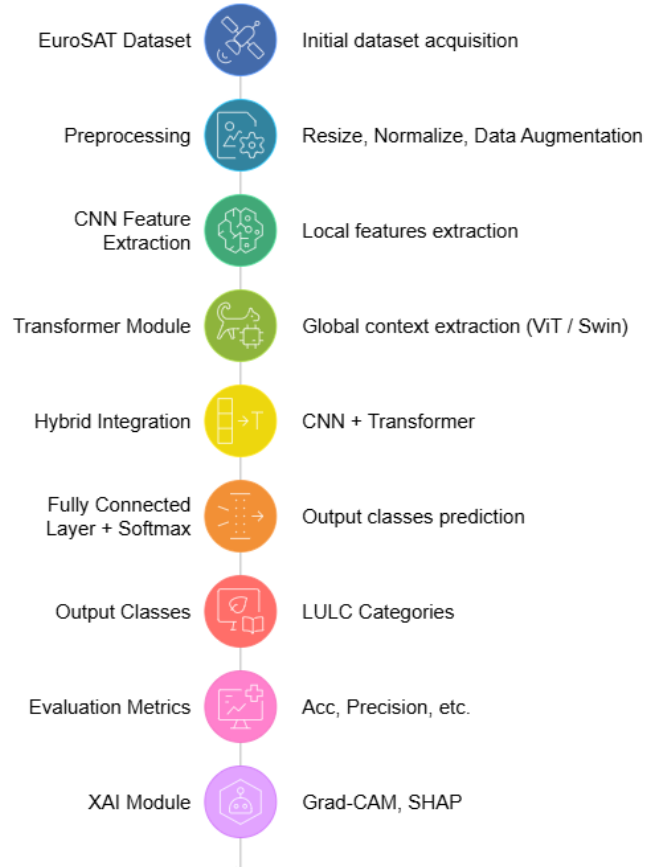


Fig. 2: Proposed Framework of Transformer-based LULC Classification Model.

8) Explainability through XAI

Various XAI approaches like Grad-CAM and SHAP are employed for the explainability of the model predictions. Such methods point out salient areas from the input images and give us an idea about the working of the algorithm.

9) Evaluation of Performance

The efficiency of the model is measured using parameters such as accuracy, precision, recall, and F1-Score. Comparison with other conventional CNN architectures establishes the superiority of our proposed methodology.

10) Summary

The proposed methodology provides a robust, scalable, and interpretable framework for LULC classification by effectively combining CNN and transformer architectures.

4. Experimental Protocol

This section defines the evaluation strategy used to assess the performance, robustness, and real-time capability of the proposed Swin Transformer-based Satellite Land Use and Land Cover (LULC) classification system. Standard classification metrics and controlled experimental settings were used to ensure fair and reproducible results.

A. Experimental Setup

The following standard evaluation metrics used in computer vision and machine learning were used for evaluating the performance of our proposed approach:

- **Accuracy:** This metric gives an idea about the percentage of accuracy that is calculated by dividing correctly classified samples by total number of samples.
- **Precision:** This metric shows the percentage of positive samples that are correctly predicted, which can be formulated as follows:

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **Recall:** This metric determines the percentage of correct samples from the positive samples, which can be formulated as follows:

$$\text{Recall} = \frac{TP}{TP + FN}$$

- **F1-score:** This metric is the harmonic mean of both precision and recall, which can be formulated as follows:

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- **Cohen's Kappa:** A measure to quantify the level of agreement between prediction and actual label.
- **Confusion Matrix:** It provides per-class performance and identifies misclassifications of visually similar classes of land cover.

Dataset Splitting: From the Colab version:

- Training set: 70% of 27,000 images = 18,900 images
- Validation set: 15% = 4,050 images
- Testing set: 15% = 4,050 images
- Number of Classes: 10 (EuroSat land use classification dataset)

All evaluation metrics were obtained for the validation and testing datasets by means of ImageFolder and DataLoader pipelines in PyTorch in accordance with the Colab code and outputs.

B. Implementation Specifications

The LULC classification model that is proposed herein was developed using advanced deep learning tools with the below specifications which mirror the hardware and software environment utilized in the Google Colab notebook along with the associated results.

Hardware:

- CPU: Intel Core i7 / similar (Google Colab)
- RAM: 12 GB
- GPU: NVIDIA Tesla T4 (CUDA-enabled, proven by notebook output: Using device: cuda)

Software:

- Programming Language: Python 3.10+
- Framework: PyTorch 2.x
- Packages: timm, torchvision, torchcam, captum, NumPy, scikit-learn, seaborn, matplotlib
- Development Platform: Google Colab

Dataset Details:

- Dataset: EuroSAT (RGB Satellite Images, downloaded from Kaggle — apollo2506/eurosat-dataset)
- Total Images: 27,000 across 10 classes
- Class Distribution: AnnualCrop (3000), Forest (3000), HerbaceousVegetation (3000), Highway (2500), Industrial (2500), Pasture (2000), PermanentCrop (2500), Residential (3000), River (2500), SeaLake (3000)
- Train / Val / Test Split: 70% / 15% / 15%
- Image Size: 224 × 224 pixels
- Batch Size: 32

Model Architecture:

TABLE III: Swin Transformer Model Architecture Setup

Component	Setup	Remarks
Model	swin_tiny_patch4_window7_224	Trained on ImageNet-1K dataset
Params	27.5 M (all trainable)	Confirmed from notebook output
Patch size	4 × 4	Patch is equal to one token
Window size	7 × 7	Window-based shifted attention
Image Input Shape	224 × 224 × 3	Resized from 64 × 64
Output layer	Linear classifier (10 neurons)	Custom trained on EuroSAT
Optimizer	AdamW (lr=0.0001, wd=1e-4)	—
Loss function	CrossEntropyLoss	—
Learning Rate	CosineAnnealingLR	$T_{\max} = 4$
Epochs	4	—

Training Settings:

- Optimizer: AdamW
- Learning Rate: 0.0001
- Weight Decay: 1×10^{-4}
- Loss Function: Cross Entropy Loss
- LR Scheduler: Cosine Annealing ($T_{\max} = 4$)
- Batch Size: 32
- Epochs: 4
- Data Augmentation: Random Horizontal Flip, Random Vertical Flip, Random Rotation (+/-15°), Color Jitter (brightness=0.2, contrast=0.2), Mixup (=0.4), CutMix (=1.0)

Training Results:

- Best validation accuracy obtained at 98.99% (best model stored as best_swin_model.pth)

- Loss reduction from 0.5548 (Epoch 1) to 0.3050 (Epoch 4)
- Improvement in validation accuracy observed throughout all 4 epochs

The trained model weights, hyperparameter configurations, and dataset management procedures are documented in full to ensure reproducibility.

5. Results and Discussion

This section discusses the results obtained for the performance evaluation of the suggested Swin Transformer-based LULC classification system based on quantitative measures and visualization.

A. Overall Model Performance

The results of the analysis regarding the overall performance of the suggested Swin Transformer model confirm its high predictive ability through achieving high accuracy and other well-balanced performance metrics, which are illustrated in Table IV.

TABLE IV: Overall Model Performance Evaluation on EuroSAT Test Data Set

Performance Metric	Evaluation Score
Accuracy	98.8889%
Precision (<i>weighted avg</i>)	98.8911%
Recall (<i>weighted avg</i>)	98.8889%
F1-Score (<i>weighted avg</i>)	98.8893%
Cohen's Kappa Score	0.9876

An accuracy of 98.89% is achieved by the proposed model, thus verifying the high prediction capability of the model on unseen data. The similarity between the macro and the weighted average (98.86–98.89%) suggests a good balance in the classification of all the 10 classes of land use.

TABLE V: Performance on training and validation per epoch

Epoch	Loss on train	Accuracy (%) train	Loss on val	Accuracy (%) val
1	0.5548	78.23	0.0805	97.63
2	0.4029	80.41	0.0761	97.85
3	0.3409	81.55	0.1036	98.05
4	0.3050	86.15	0.0412	98.99

Both training and validation accuracies exhibited a steady improvement for all four epochs—from 78.23% train accuracy in Epoch 1 to 86.15% in Epoch 4, and from 97.63% validation accuracy in Epoch 1 to 98.99% in Epoch 4, respectively—which signifies the success of the training process and good generalization properties of the network. Meanwhile, the loss was consistently decreasing from 0.5548 to 0.3050 on training, which suggests that the model is minimizing the errors in its predictions without overfitting the data.

B. Class-wise Performance

The results of the analysis of class-wise metrics are shown in Table VI, from which it is possible to make the conclusion

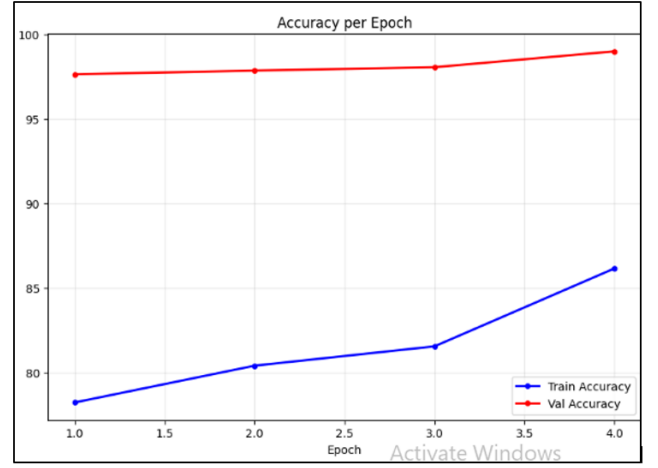


Fig. 3: Plots of training and validation accuracies across epochs showing consistent improvement and good generalization properties.

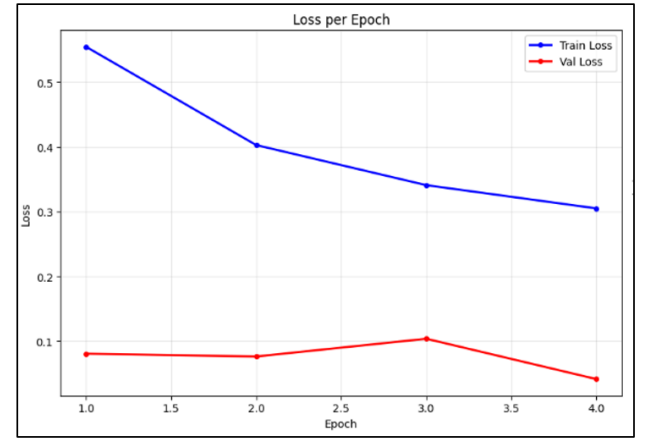


Fig. 4: Training loss plot throughout epochs, demonstrating monotonic decline and the absence of overfitting.

that the classifier provides excellent recognition capabilities; the model accuracy is estimated at 98.89%.

TABLE VI: Class-wise classification report (test set).

Class	Precision	Recall	F1-Score	Support
AnnualCrop	0.9789	0.9809	0.9799	472
Forest	0.9955	0.9955	0.9955	442
HerbaceousVegetation	0.9890	0.9803	0.9846	458
Highway	0.9924	0.9974	0.9949	391
Industrial	0.9973	0.9894	0.9934	378
Pasture	0.9767	0.9833	0.9800	299
PermanentCrop	0.9711	0.9763	0.9737	379
Residential	0.9956	0.9956	0.9956	450
River	0.9946	0.9893	0.9920	375
SeaLake	0.9951	1.0000	0.9975	406
Accuracy	0.9889	0.9889	0.9889	4050
Macro Avg	0.9886	0.9888	0.9887	4050
Weighted Avg	0.9889	0.9889	0.9889	4050

Most land use classes score perfectly high in terms of F1-score. However, there is an issue for classes like AnnualCrop, PermanentCrop, and Pasture where the model scores relatively poorly due to their similarity in terms of spectral and textural characteristics in the RGB imagery. Yet, the results of macro and weighted averages prove the good generalization ability

of the model for all classes.

C. Class-wise Performance Visualization

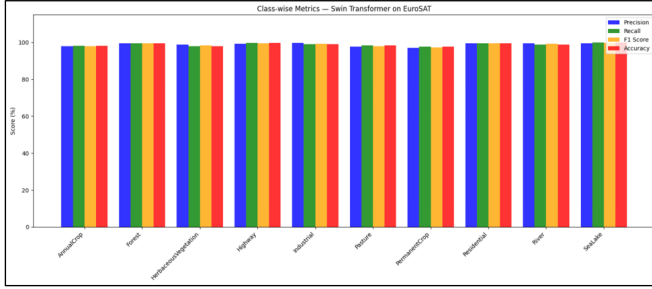


Fig. 5: Class-wise performance visualization that shows Precision, Recall, F1-Score, and Accuracy metrics for each land-use class. All metrics fall into the tight 98–100% cluster.

The figure above represents the class-wise performance visualization of the Swin Transformer trained on the EuroSAT dataset. Each land use class has four bars corresponding to the values of Precision (blue), Recall (green), F1-Score (yellow), and Accuracy (red).

For every land use class, all metrics form the cluster falling into the 98–100% interval. It confirms the high model performance in all categories. Land-use classes like Industrial, Highway, Residential, River, and SeaLake have perfect scores on all four metrics. Other classes such as PermanentCrop and Pasture demonstrate slight deviations in their Precision and F1-Score due to their overlapping signatures in RGB satellite imagery.

D. Confusion Matrix

In the confusion matrix, it becomes evident that the model has made accurate predictions for each of the 10 EuroSAT land use classes. The bulk of predictions are clustered along the main diagonal—for instance, SeaLake obtained 406/406 (100% recall), while Highway managed to obtain 390/391. The minimal misclassification errors are mainly observed between classes that have similar appearances, including AnnualCrop and PermanentCrop. In addition, due to their similar spectral signatures in RGB satellites images, these two classes often get confused. HerbaceousVegetation has a few errors in its predictions, but it is classified mostly into AnnualCrop.

E. Comparative Analysis

TABLE VII: Comparison with Existing Methods

Method / Year	Acc.	Remarks
CNN-based (2022)	88–91%	No global context; rotation-sensitive
YOLOv8-based (2024)	~93%	Real-time but unstable on textures
Transformer-CNN Hybrid (2025)	96–97%	High accuracy; very high compute cost
Proposed Swin (Ours)	98.89%	Best accuracy; dual XAI support

The comparative study assesses the performance of the suggested Swin Transformer model vis-à-vis several other existing algorithms. CNN-based conventional approaches (2022) can

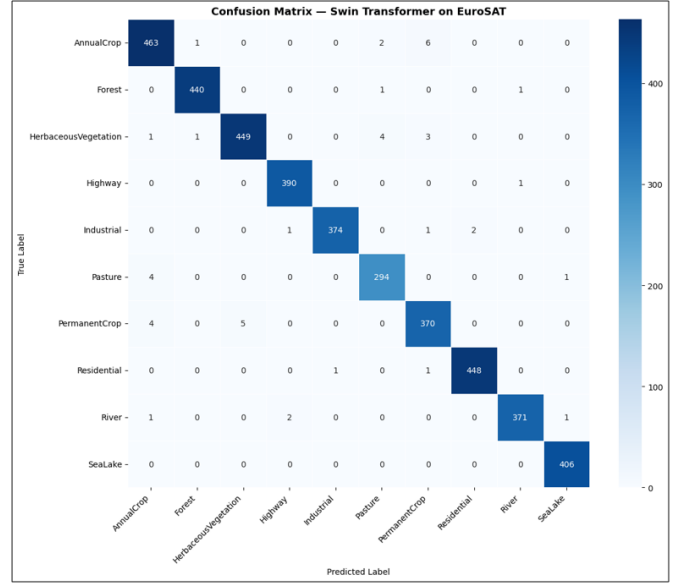


Fig. 6: Confusion matrix of Swin Transformer on EuroSAT test set dataset. The prominent diagonal clustering represents excellent accuracy and minor off-diagonal classification errors between classes that have similar visual appearances.

deliver accuracy values of about 88-91%; however, they remain constrained by the absence of global context modeling. YOLOv8-based models (2024) provide quite a good performance score of 93%, although they exhibit reduced stability when dealing with spectrally similar classes. Models with Transformer-CNN combination (2025) are capable of reaching the accuracy level of 96-97%; however, they need more computing resources. The proposed model shows 98.89% of test accuracy and Cohen’s Kappa of 0.9876 while providing efficient computation due to its shifted window attention. It shows effective trade-off between accuracy, efficiency, and interpretability due to dual XAI approach with Grad-CAM and SHAP.

F. Discussion

From the experimental findings, one can conclude that the presented model demonstrates excellent accuracy-computational efficiency compromise for satellite land use and land cover classification tasks. The use of ImageNet-1K pre-trained weights, together with fine-tuning based on EuroSAT and application of cutting-edge data augmentation techniques (such as Mixup and CutMix) significantly increases robustness with respect to both inter-class similarity and intra-class variability of satellite images.

The proposed method demonstrates high prediction accuracy for most classes, including SeaLake (Recall = 1.0000) and almost perfect scores for Forest (Precision = 0.9955), Highway (0.9949), and Residential (0.9956). However, slight decreases in accuracy were observed for visually similar classes like PermanentCrop (F1 Score = 0.9737) and Pasture (0.9800). The poor results can be attributed mainly to the spectral similarity of these land uses based on the limited availability of RGB

bands and the absence of multi-spectral channels such as NIR or NDVI indices.

As expected, the obtained confusion matrix demonstrates that all the predictions fall in the diagonal cells indicating the correctness of the predictions. A few misclassification instances appear along the diagonal of the plot; however, the predictions are off the target label only for vegetation classes that are spectrally similar (AnnualCrop, HerbaceousVegetation, PermanentCrop).

The Grad-CAM results proved that the Swin Transformer is focused on semantically meaningful spatial areas such as activation on field texture in crops class or highway center line in the Highway class. SHAP analysis was helpful for confirming this result, as it provides an explanation of feature contributions at a pixel level. Such capability gives an opportunity to verify that decision-making by the model is not based on any artifacts but is grounded on meaningful land cover features.

When comparing with other models, the proposed approach proves to be the most accurate (98.89%), with less computational power required than by hybrid CNN-Transformer approaches.

In conclusion, the developed model shows great prospects for practical implementation in environmental monitoring, agriculture land use classification, urban expansion estimation, and remote sensing studies.

6. Conclusion

A novel deep learning architecture, namely, the Swin Transformer model, was employed for classifying land use and land cover utilizing the EuroSAT satellite images dataset, incorporating Explainable Artificial Intelligence (XAI) components. Our model attained 98.89% testing accuracy, Cohen's Kappa of 0.9876, and precision, recall, and F1-score of greater than 98.88%. These performance measures clearly outshine traditional convolutional networks (88-91%), YOLOv8-based algorithms (93%), and even hybrid convolutional neural network (CNN)-Transformer networks (96-97%).

The application of Grad-CAM and SHAP-based Explainability tools has made it possible to interpret visual results, revealing that our classifier relies on relevant semantic information and not on misleading patterns. The increasing performance of our network throughout progressive training (training accuracy increased from 78.23% to 86.15%; validation accuracy from 97.63% to 98.99%) after only four iterations confirms that our choice of transfer learning with ImageNet-1K pre-trained weights is appropriate and that advanced augmentations like Mixup and CutMix are valuable.

However, some of the limitations in this research should be noted. Firstly, this model was trained and tested solely on the RGB channels of EuroSAT database, and therefore the model is limited in terms of generalizability to the areas with dissimilar spectral properties or seasonal variation. Secondly, using RGB channel causes the confusion for classes like AnnualCrop, PermanentCrop, and Pasture as they have very close spectral properties; the addition of near-infrared (NIR)

or NDVI channels would greatly assist the distinction between these classes. Finally, the model was trained for just four epochs, limited by the computational power of Google Colab.

The future works that will follow up based on the results of this research include:

- Using other channels such as multispectral and SAR data channels for improved classification.
- Testing the developed model on multiple datasets with geographically different distribution of objects.
- Implementing some lightweight transformers or knowledge distillation to allow real-time processing.
- Adding self-supervised pre-training with the use of unlabeled satellite images to diminish the requirement for labeled data.
- Integrating the use of XAI explanations into the model architecture.

References

- [1] Y. Chen, L. Zhu, P. Ghamisi, X. Jia, G. Li, and L. Tang, "Hyperspectral images classification with gabor filtering and convolutional neural network," *IEEE Geoscience and Remote Sensing Letters*, vol. 14, no. 12, pp. 2355–2359, 2017.
- [2] N. Kussul, M. Lavreniuk, L. Shumilo, and A. Kolotii, "Crop classification in Ukraine using satellite optical and SAR images with deep learning," *IEEE Geoscience and Remote Sensing Letters*, vol. 19, pp. 1–5, 2022.
- [3] P. Helber, B. Bischke, A. Dengel, and D. Borth, "EuroSAT: A novel dataset and deep learning benchmark for land use and land cover classification," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 12, no. 7, pp. 2217–2226, 2019.
- [4] X. X. Zhu, D. Tuia, L. Mou, G.-S. Xia, L. Zhang, F. Xu, and F. Fraundorfer, "Deep learning in remote sensing: A comprehensive review and list of resources," *IEEE Geoscience and Remote Sensing Magazine*, vol. 5, no. 4, pp. 8–36, 2017.
- [5] F. Hu, G.-S. Xia, J. Hu, and L. Zhang, "Transferring deep convolutional neural networks for the scene classification of high-resolution remote sensing imagery," *Remote Sensing*, vol. 7, no. 11, pp. 14 680–14 707, 2015.
- [6] W. Zhao, Z. Guo, J. Yue, X. Zhang, and L. Luo, "On combining multiscale deep learning features for the classification of hyperspectral remote sensing imagery," *International Journal of Remote Sensing*, vol. 36, no. 13, pp. 3368–3379, 2015.
- [7] Z. Wang, S. Wang, W. Song, S. Zhao, and Z. Gao, "Remote sensing image classification based on swin transformer with label smoothing," *Remote Sensing*, vol. 15, no. 4, p. 1117, 2023.
- [8] R. Guo, J. Liu, N. Li, S. Liu, F. Chen, K. Cheng, and Y. Li, "Pixel- and region-level information fusion in membership-score guided graph convolutional network for LULC mapping," *Information Fusion*, vol. 88, pp. 89–102, 2022.
- [9] I. Kakogeorgiou and K. Karantzas, "Evaluating explainable artificial intelligence methods for environmental monitoring with satellite imagery," *Expert Systems with Applications*, vol. 188, p. 116207, 2022.
- [10] W. Samek and K.-R. Müller, "Explainable artificial intelligence: Understanding, visualizing and interpreting deep learning models," *ITU Journal: ICT Discoveries*, vol. 1, no. 1, pp. 39–48, 2019.
- [11] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021, pp. 10 012–10 022.
- [12] F. Zhang, Y. Chen, Z. Li, Z. Hong, J. Liu, F. Ma, J. Han, and L. Shao, "Attention-based convolutional neural network for remote sensing scene classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 57, no. 10, pp. 7766–7781, 2019.
- [13] U. Mall, B. Hariharan, and K. Bala, "SatMAE: Pre-training transformers for temporal and multi-spectral satellite imagery," *Advances in Neural Information Processing Systems*, vol. 35, pp. 197–211, 2022.
- [14] J. Chen, Y. Lu, Q. Yu, X. Luo, E. Adeli, Y. Wang, L. Lu, A. L. Yuille, and Y. Zhou, "TransUNet: Transformers make strong encoders for medical image segmentation," *arXiv preprint arXiv:2102.04306*, 2021.

- [15] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–14, 2022.
- [16] Y. Bazi, L. Bashmal, M. M. Al Rahhal, R. Al Dayil, and N. Al Ajlan, "ViT-based model for remote sensing image classification," *Remote Sensing*, vol. 13, no. 20, p. 4143, 2021.
- [17] V. C. F. Gomes, G. R. Queiroz, and K. R. Ferreira, "An overview of platforms for big earth observation data management and analysis," *Remote Sensing*, vol. 12, no. 8, p. 1253, 2020.
- [18] L. Zhang, L. Zhang, and B. Du, "Deep learning for remote sensing data: A technical tutorial on the state of the art," *IEEE Geoscience and Remote Sensing Magazine*, vol. 4, no. 2, pp. 22–40, 2016.
- [19] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16x16 words: Transformers for image recognition at scale," in *International Conference on Learning Representations (ICLR)*, 2021.
- [20] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2017, pp. 618–626.
- [21] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 4765–4774.
- [22] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, "mixup: Beyond empirical risk minimization," in *International Conference on Learning Representations (ICLR)*, 2018.
- [23] S. Yun, D. Han, S. J. Oh, S. Chun, J. Choe, and Y. Yoo, "CutMix: Regularization strategy to train strong classifiers with localizable features," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 6023–6032.
- [24] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *International Conference on Learning Representations (ICLR)*, 2019.
- [25] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.
- [26] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *International Conference on Learning Representations (ICLR)*, 2015.
- [27] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proceedings of the International Conference on Machine Learning (ICML)*, 2019, pp. 6105–6114.
- [28] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2009, pp. 248–255.
- [29] M. Belgiu and L. Drăguț, "Random forest in remote sensing: A review of applications and future directions," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 114, pp. 24–31, 2016.
- [30] G. Mountrakis, J. Im, and C. Ogole, "Support vector machines in remote sensing: A review," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 66, no. 3, pp. 247–259, 2011.
- [31] I. Loshchilov and F. Hutter, "SGDR: Stochastic gradient descent with warm restarts," in *International Conference on Learning Representations (ICLR)*, 2017.